

Jose Sanchez Gonzalez

Robotics / Machine Learning Engineer

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Summary

M.S. Artificial Intelligence building robotics software from perception to actuation. Work includes a gated UR5e simulation loop, ROS 2 recorded replay, C++ sensor geometry, and Quest VR control of physical Berkeley Humanoid Lite arms.

Technical Skills

Languages: Python, C++17, NumPy, PyTorch

Robot Integration: Inverse kinematics, SocketCAN, gravity feedforward, WebXR, ROS 2 Humble

Perception and Simulation: RGB-D, OpenCV, time-of-flight geometry, Isaac Sim, MuJoCo

Engineering: pytest, gtest, GitHub Actions, Docker, FastAPI, Linux, Git, Slurm

Projects

Berkeley Humanoid Lite: Quest VR Teleoperation | Quest 2, WebXR, Python, inverse kinematics, SocketCAN

- Built Quest 2/WebXR teleoperation for simulated and physical Berkeley Humanoid Lite arms: Python/UDP pose bridge, inverse kinematics, and SocketCAN actuation.
- Integrated gravity feedforward, grip-based motion gating, and headset calibration; added camera, relative-depth/optical-flow, lidar, and IMU views.

Vision-Guided Pruning and ROS 2 Replay | Python, C++17, Isaac Sim, OpenCV, ROS 2 Humble

- Built RGB-D servoing with dual-ToF/contact gates for a simulated UR5e; depth-aware tracking passed 4/4 targeted low-sun trials while retaining occlusion stops.
- Reproduced 199/199 recorded decisions in ROS 2 software-in-the-loop; the C++17 ToF port matched Python on 232 recorded zones within 1e-9 m.

SPUR: Metric Depth for Robotic Pruning | PyTorch, DINOv2, Blender, FastAPI, ONNX

- Recorded 0.0445 ± 0.0057 m depth RMSE with three stereo pairs on synthetic orchard data (five seeds; stored best-validation scores).
- Packaged Docker/FastAPI inference; measured the PyTorch refiner at 156 ms p50 (V100, fp16, six precomputed-depth views); checked separate ONNX export parity to $1.53e-5$.

Humanoid Robustness Ladder | Isaac Lab, MuJoCo, PPO, ONNX, Python

- Automated PPO export and MuJoCo sim-to-sim scoring through the upstream Berkeley Humanoid Lite controller, with matched controls and scenario-linked artifacts.

Bimanual Garment Folding Evaluation | SmolVLA, PyTorch, Isaac Sim, PhysX cloth

- Integrated SmolVLA with PhysX cloth, three camera views, and the LeHome checker; separated policy rollouts from demonstration replay.

Experience

Graduate Teaching Assistant, Oregon State University | Sep 2025 – Jun 2026

- Led 2-hour Operating Systems (C++) and Computational Methods (COMSOL) studios for 25+ students; taught debugging, concurrency, and numerical simulation.

Math and Computer Science Tutor, Oregon State University | Nov 2022 – Jun 2025

- Tutored 50+ students in mathematics and computer science; adapted debugging guidance and technical explanations to individual learning needs.

Education

Oregon State University, Corvallis, OR

Master of Science in Artificial Intelligence | Jun 2026

Bachelor of Science in Computer Science (Applied: Data Science); Mathematics Minor | Jun 2025